

Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

Student Name · Enrollment Number · Semester VII · Information Technology
· College Name



The Problem With Finance Today

Privacy at Risk

Centralized databases expose user data to breaches, surveillance, and third-party exploitation.

Always Online

Internet dependency makes every transaction fragile — no connectivity, no commerce.

Costly Infrastructure

Monthly POS fees and cloud hosting create barriers for small merchants and local businesses.



Three Apps. One Ecosystem.

Spendly

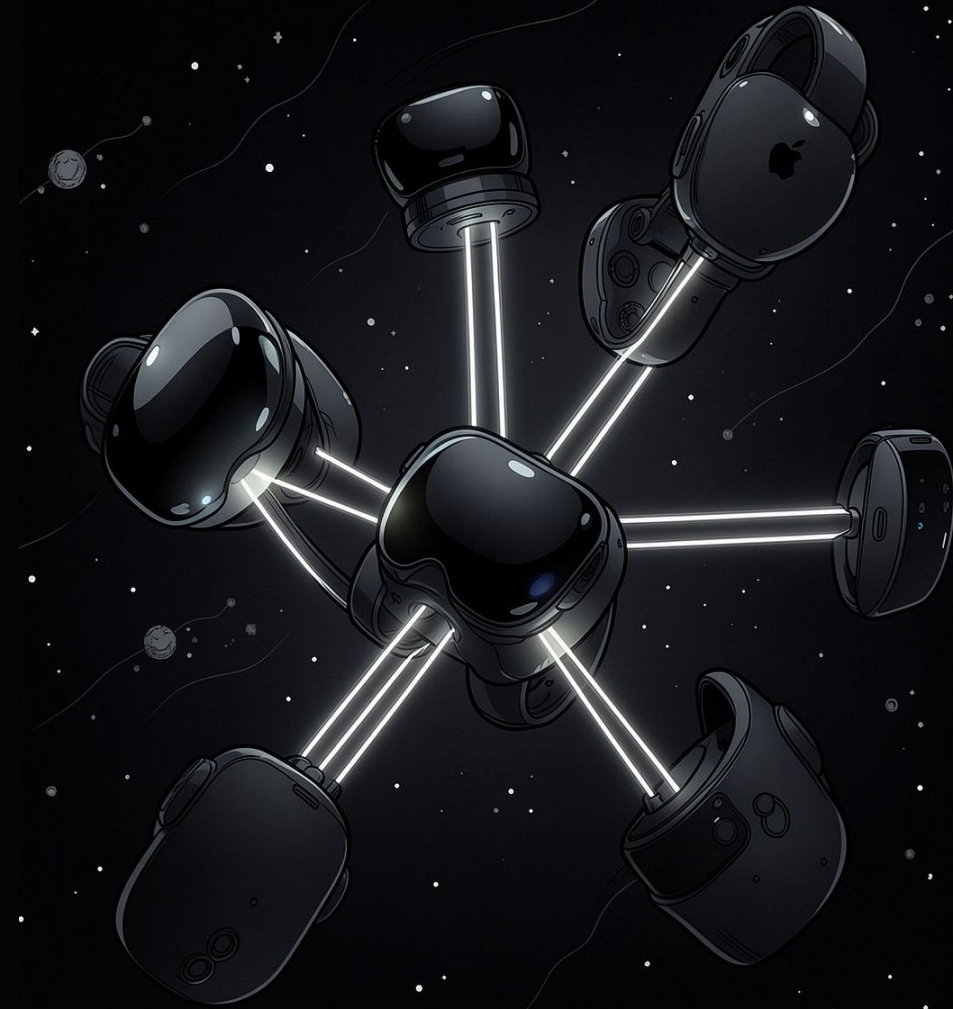
The consumer wallet. Track expenses, claim offers, pay merchants — entirely offline.

Spendly Shop

The merchant terminal. Inventory, billing, and analytics without a server.

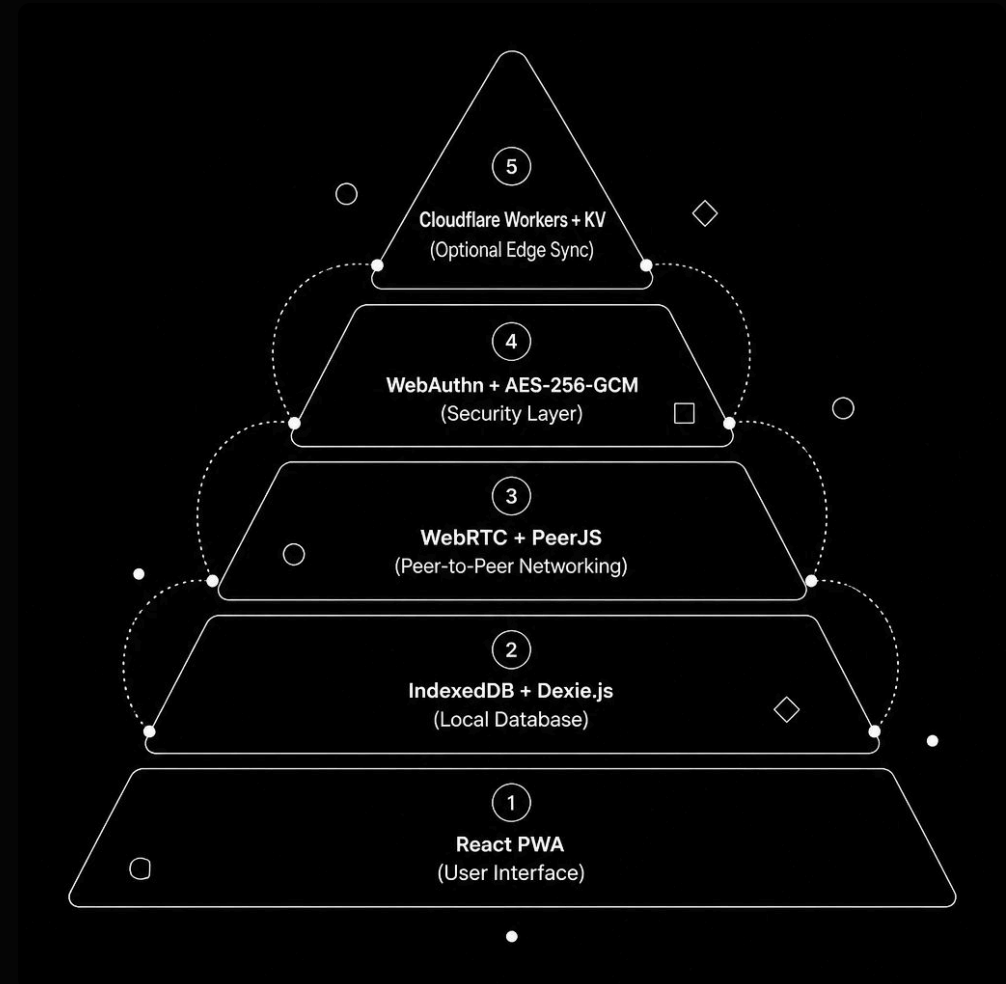
Spendly Radar

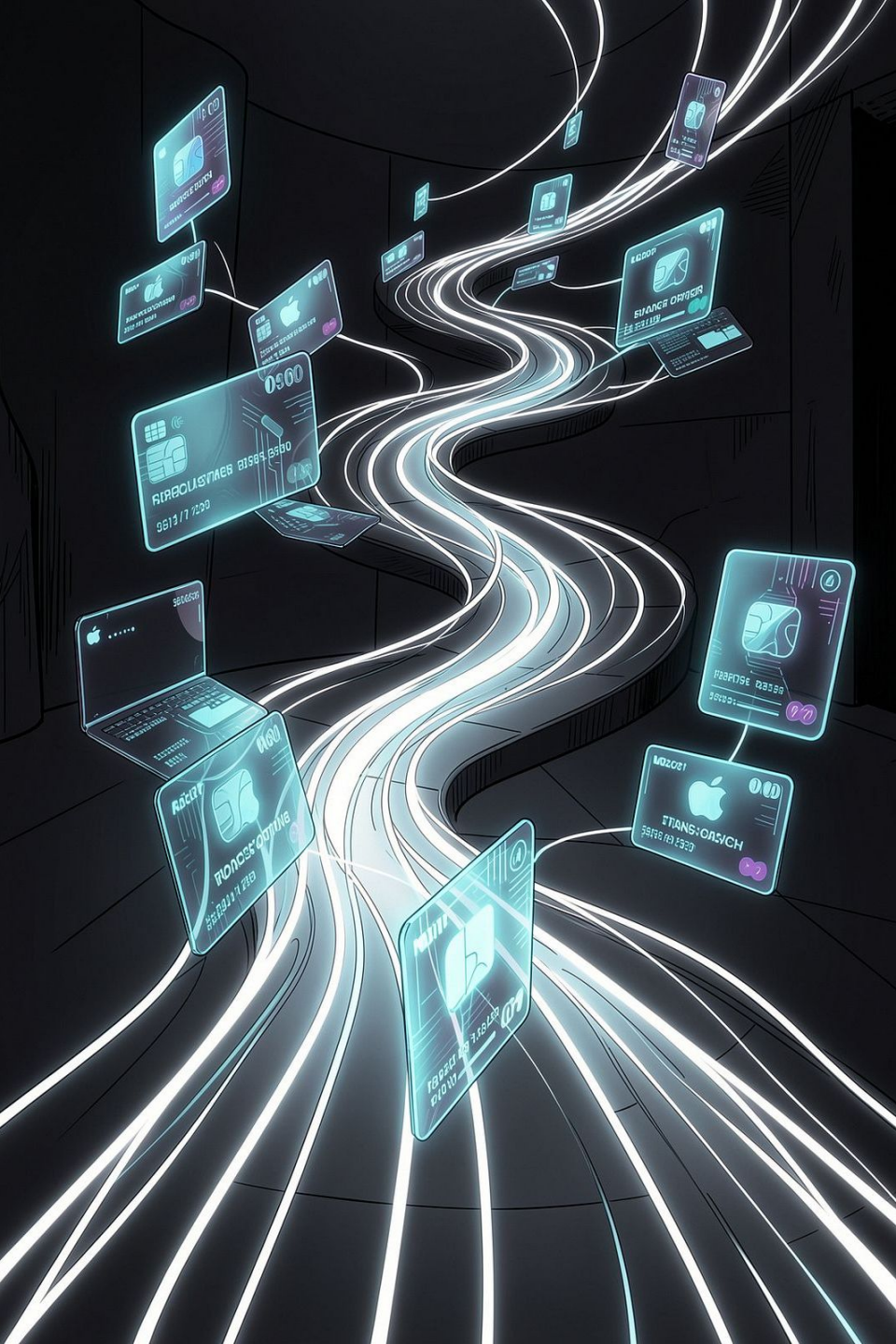
The discovery layer. Find nearby offers and merchants through encrypted peer-to-peer signals.



System Architecture

A fully offline-first stack — every layer operates locally, synchronizing peer-to-peer only when needed. No central server. No cloud dependency. Zero-knowledge by design.





How a Transaction Flows

01

Offer Discovery

Radar detects nearby merchant offers via encrypted peer-to-peer signals.

02

Coupon Claim & Scan

User claims offer, scans barcode — all processed locally on device.

03

Validation & Invoice

Product lookup, offer validation, and invoice generation happen offline.

04

Share & Analyze

Invoice shared peer-to-peer; expense saved and analytics updated instantly.

Technology Stack

Frontend

React · PWA · Responsive Design

Database

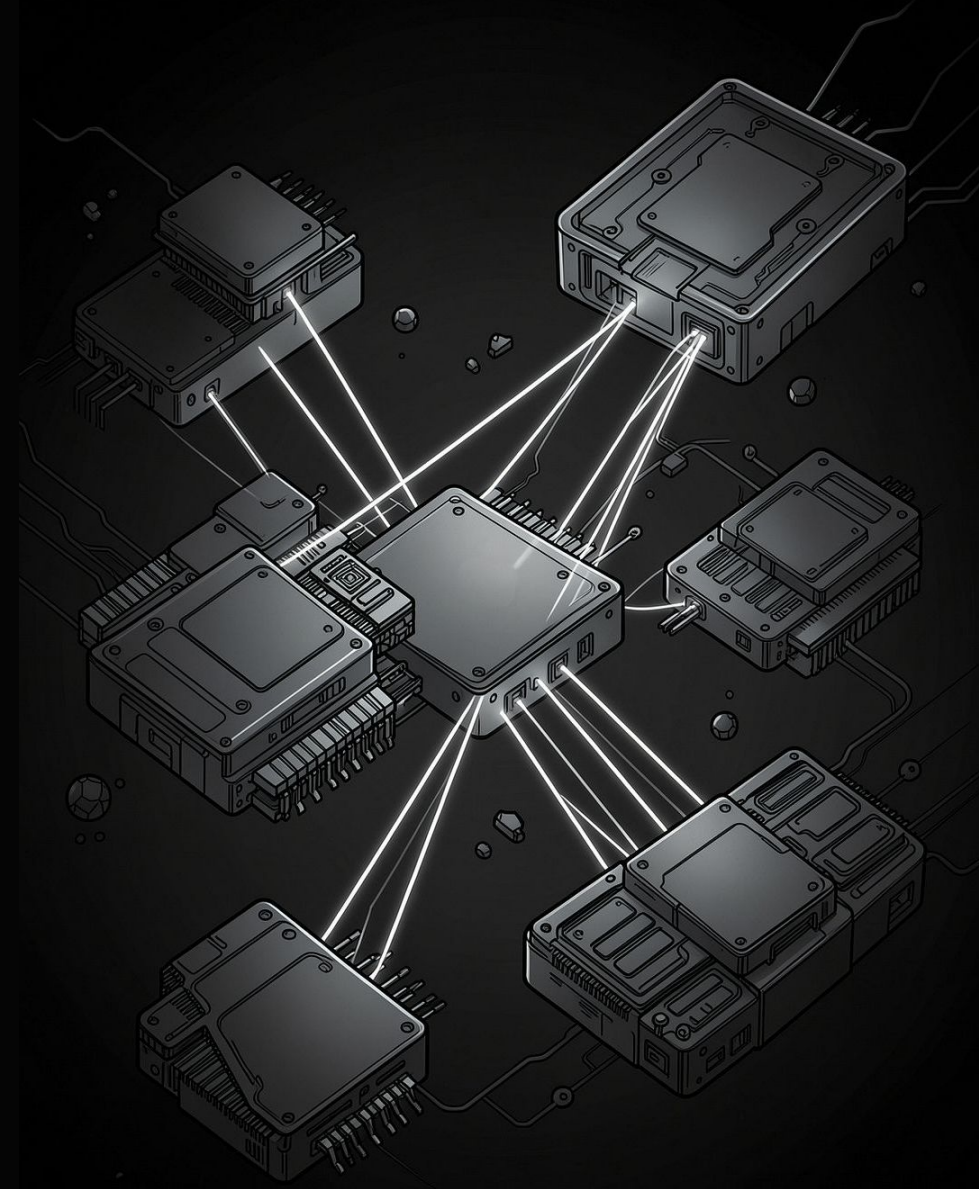
IndexedDB · Dexie.js · Local-First Storage

Networking

WebRTC · PeerJS · Cloudflare Workers

Security

WebAuthn · AES-256-GCM · Zero-Knowledge Proofs



Security by Architecture

Zero-Knowledge

No transaction data ever leaves the device unencrypted.
Merchants see only what is required.

Offline-First

All data resides in IndexedDB. No cloud breach surface. No third-party exposure.

Hardware-Grade Auth

WebAuthn + Biometric Unlock + AES-256-GCM encryption — the same standard as Apple Secure Enclave.



Features That Redefine Commerce



Expense & Cash Wallet

Real-time tracking of every rupee – offline, private, instant.



Merchant Billing & QR Invoice

Generate, share, and settle invoices without internet or hardware.



Inventory & EMI Alerts

Smart stock management with automated payment reminders.



Offline Sync & Analytics

Deep insights that sync silently when connectivity is restored.



Advantages & What Comes Next

Core Advantages

- **Zero Server Cost** — No cloud infrastructure, no monthly fees.
- **True Privacy** — Data never leaves the device without consent.
- **Fast Billing** — Peer-to-peer transactions in milliseconds.
- **Offline-First** — Works without internet, everywhere.

Future Scope

AI-Powered Spending
Insights

Voice-Activated Billing

UPI & NFC Integration

GST Compliance & Multi-
Store

Spendly Finance. Private. Offline. Yours.

A new standard for peer-to-peer commerce — built for a world that demands privacy, resilience, and elegance.

Thank You · Questions Welcome

Prrrpl Finance Ecsctvacom

Örmpete Offline-First Filna Xivannes



Made with GAMMA